Bonus 1: Test-Day Quick Reference Sheet

These two pages condense the whole course into the facts, decisions, and traps that matter most on examination day. Read them the night before and again on the morning of the examination. They are a summary, not a substitute for the chapters: if any line is unfamiliar, return to the chapter shown in brackets.

Essentials at a Glance

Topic	What to remember
Order of operations	Grouping → exponents → × and ÷ left to right → + and − left to right. A fraction bar groups. [Ch. 1]
Signs	$-3^2 = -9$ but $(-3)^2 = 9$. A negative outside parentheses changes every sign inside. [Ch. 1, 3]
Equations	Same operation on both sides; undo in reverse order. False result: no solution. True result: all real numbers. [Ch. 4]
Inequalities	Reverse the sign when multiplying or dividing by a negative. Open circle for $< >$; closed for $\leq \geq$. [Ch. 5]
Functions	One output per input. f(a): replace every x with (a). Vertical line test. [Ch. 6]
Slope	$m = (y_2 - y_1)/(x_2 - x_1)$. Horizontal: 0. Vertical: undefined. Parallel: equal slopes. [Ch. 7]
Line forms	$y = mx + b$; $y - y_1 = m(x - x_1)$; $Ax + By = C$. Intercepts: set the other variable to 0. [Ch. 7, 8]
Systems	One point (different slopes), none (parallel), infinitely many (same line). Check in both equations. [Ch. 9]
Inequality graphs	Dashed for $< >$, solid for $\leq \geq$; shade using a test point such as (0, 0). [Ch. 10]
Exponents	Multiply: add exponents. Divide: subtract. Power of a power: multiply. $a^0 = 1$; $a^{-n} = 1/a^n$. [Ch. 11]
Exponential	$y = a \cdot b^x$. Growth $b = 1 + r$; decay $b = 1 - r$. Constant ratios in tables. [Ch. 12]
Sequences	Arithmetic $a_n = a_1 + (n - 1)d$. Geometric $a_n = a_1 \cdot r^{n-1}$. [Ch. 13]
Polynomials	$(a \pm b)^2 = a^2 \pm 2ab + b^2$; $(a + b)(a - b) = a^2 - b^2$. Subtracting changes every sign. [Ch. 14]
Factoring	GCF first, then count terms; check by multiplying. Sums of squares do not factor. [Ch. 15]
Parabolas	Axis $x = -b/(2a)$. $a > 0$ opens up (minimum); $a < 0$ opens down (maximum). Vertex form: vertex (h, k). [Ch. 16]
Quadratic equations	$x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$. Discriminant > 0: two; $= 0$: one; < 0: none. Remember ±. [Ch. 17]
Statistics	Mean uses every value; median resists outliers. $IQR = Q_3 - Q_1$. Correlation is not causation. [Ch. 18]

Which Method Should I Use?

If the problem looks like...	Do this
A linear equation with fractions	Multiply every term by the least common denominator first.
A system with a variable already isolated	Substitution.
A system in standard form	Elimination: make one pair of coefficients opposites, then add.
A quadratic with no x-term, such as $3x^2 = 48$	Isolate x^2 and take ± square roots.
A quadratic that factors easily	Standard form, factor, zero product property.
A quadratic that does not factor	Quadratic formula (find the discriminant first).
"Maximum," "minimum," "highest," "greatest area"	Find the vertex: $x = -b/(2a)$, then substitute.
"When does it hit the ground?"	Set the height equal to 0 and solve; reject negative times.
A table of values	Constant differences: linear. Constant ratios: exponential. Constant second differences: quadratic.
"At least" / "at most" / "no more than"	Write an inequality: ≥ / ≤ / ≤.

Twelve Traps to Avoid

#	Trap
1	Forgetting to reverse an inequality after dividing by a negative number.
2	Distributing a negative to the first term only: $-2(x - 5) = -2x + 10$, not $-2x - 10$.
3	Writing $(x + 4)^2$ as $x^2 + 16$; the middle term 8x is missing.
4	Computing slope as run over rise, or subtracting coordinates in different orders.
5	Reading the vertex of $y = (x + 3)^2 - 1$ as (3, −1); it is (−3, −1).
6	Dropping the ± when taking a square root.
7	Using 0.06 instead of 1.06 for 6% growth, or 0.06 instead of 0.94 for 6% decay.
8	Adding exponents when combining like terms: $x^2 + x^2 = 2x^2$, not x^4.
9	Using the zero product property when the product is not zero.
10	Finding x in a system and forgetting to find y.
11	Ordering data incorrectly, or not at all, before finding the median.
12	Answering with the wrong quantity or without units in a word problem.